AmarVote

Secure Cryptographic Voting Platform

Complete System Guide

This guide covers the full end-to-end workflow of AmarVote:

- Platform Setup & Authentication
- Election Creation & Configuration
- Guardian Key Ceremony
- Encrypted Voting & Benaloh Challenge
- Tallying & Cryptographic Decryption
- Independent Ballot Verification

ElectionGuard Powered | End-to-End Verifiable

Version 1.0 | April 17, 2026

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About This Guide

This document is the authoritative, comprehensive reference for the **AmarVote** platform. It is written for a general audience — no prior knowledge of cryptography, election administration, or software engineering is required.

AmarVote is built on the **ElectionGuard** specification developed by Microsoft, which guarantees *end-to-end verifiability*: every voter can independently confirm that their ballot was cast as intended, recorded as cast, and counted as recorded — without ever exposing how they voted.

> **Note**
>
> This guide deliberately mirrors the actual workflow of the application in the exact order you would encounter it. Read sequentially for first-time use, or jump to a specific chapter if you need a quick reference.

Roles at a Glance

Role	Responsibilities
Owner	Full platform control: manages the authorised-user list, assigns roles, and sets global permissions.
Admin	Creates and administers elections, configures election parameters, and oversees the tally phase.
Guardian	Participates in the cryptographic key ceremony; holds a share of the decryption key needed to tally results.
Voter	Registers, receives election access, casts an encrypted ballot, and independently verifies the result.

> **Note**
>
> **Permission hierarchy:** Owner > Admin > User (voter). The Owner can elevate or restrict any account. All roles require a registered account.

Part I

Platform Foundation

Chapter 1

Access Control & Authorisation System

AmarVote is a closed platform — not just anyone can register. Before a user can even create an account, their email address must appear on the **Authorised User List** managed by the Owner. This design ensures that elections are only accessible by vetted participants.

1.1 The Authorised User List

The Owner maintains a registry of permitted email addresses. Each entry in the list specifies:

- The user's `email address`
- The `assigned role` (Owner, Admin, or User)
- Whether the user `can create elections` (a permission that can be granted independently of role)

1.1.1 Adding Individual Users

1. Log in to AmarVote with an Owner account.
2. Navigate to **Authenticated Users** in the sidebar.
3. Click **Add User**.
4. Enter the email address in the `Email` field.
5. Select the appropriate role from the `Role` dropdown.
6. Toggle `Can Create Elections` if applicable.
7. Click **Save**.

Tip

If you are adding a small number of users (fewer than 10), the individual entry method is faster. For larger groups, use CSV bulk upload.

1.1.2 Bulk Upload via CSV

For adding many users at once:

1. Prepare a `.csv` file with the following columns:
   `email, role, can_create_elections`
   Example row: `alice@example.com, admin, true`
2. Navigate to **Authenticated Users** → **Import CSV**.
3. Click **Choose File** and select your prepared CSV.
4. Review the preview table that appears — verify counts and roles.
5. Click **Confirm Import**.

> **Important**
> Duplicate email entries in the CSV will be ignored. If an email already exists in the system, its role will be **updated** to the value in the CSV, not duplicated.

1.2 Role Capabilities Summary

Capability	Owner	Admin	User
Manage authorised-user list	✓	✗	✗
Assign roles	✓	✗	✗
Create elections (default)	✓	✓	✗
Create elections (if granted)	✓	✓	✓
Vote in elections	✓	✓	✓
View public elections	✓	✓	✓
View private elections (if invited)	✓	✓	✓

> **Note**
> **Election creation setting:** The Owner can enable a global setting allowing *all* users (not just admins) to create elections. When this setting is active, both Admins and regular Users will see the **Create Election** button in their menu.

Chapter 2

Account Registration

Before using any feature of AmarVote, every participant — voter, guardian, or admin — must create a personal account. The registration process involves email verification and optional two-factor authentication setup.

2.1 Prerequisites

Important

You can only register if your email address has been added to the Authorised User List by an Owner. If you attempt to register with an unlisted email, the system will reject your request.

2.2 Step-by-Step Registration

2.2.1 Step 1 — Navigate to Registration

1. Open your web browser and go to the AmarVote application URL.
2. On the landing page, click **Register** or navigate to `/register`.

2.2.2 Step 2 — Submit Your Email Address

1. Enter your `Email Address` in the provided field. This must exactly match the email your Owner added to the authorised list.
2. Click **Send Verification Code**.
3. The system will send a **One-Time Password (OTP)** to your email inbox.

Tip

Check your spam or junk folder if you do not receive the email within two minutes. The OTP is typically valid for **10 minutes**.

2.2.3 Step 3 — Verify Your Email

1. Open the email from AmarVote.

2. Copy the 6-digit verification code.
3. Return to the registration page and enter the code in the `Verification Code` field.
4. Click **Verify Email**.
5. If successful, you will be advanced to the password setup step.

2.2.4 Step 4 — Set Your Password

1. Enter a strong password in the `Password` field. Recommended: at least 12 characters, including uppercase letters, numbers, and symbols.
2. Re-enter the same password in the `Confirm Password` field.
3. Click **Create Account**.

> **Warning**
>
> **Never share your password.** AmarVote staff will never ask for your password. Store it in a secure password manager.

2.3 Two-Factor Authentication (2FA) — Optional but Recommended

After your account is created, you can enable Two-Factor Authentication using **Google Authenticator** (or any TOTP-compatible app such as Authy or Microsoft Authenticator).

2.3.1 Enabling 2FA

1. Log in to your new account.
2. Click your profile avatar in the top-right corner and select **Profile Settings**.
3. Scroll to the **Security** section and click **Enable Two-Factor Authentication**.
4. A QR code will be displayed on screen.
5. Open the Google Authenticator app on your smartphone.
6. Tap the + button and select **Scan a QR code**.
7. Point your camera at the QR code on screen.
8. Google Authenticator will display a 6-digit rotating code. Enter this code in the `Verification Code` field on AmarVote.
9. Click **Confirm** to activate 2FA.

Important

Back up your 2FA recovery codes! AmarVote will display one-time recovery codes after 2FA is enabled. Print or save them somewhere safe. If you lose access to your authenticator app and have no recovery codes, you may be permanently locked out of your account.

Chapter 3

Logging In

3.1 Standard Login

1. Navigate to the AmarVote application and click **Login** or go to `/login`.
2. Enter your registered `Email Address`.
3. Enter your `Password`.
4. Click **Sign In**.

3.2 Login with Two-Factor Authentication

If you have 2FA enabled:

1. Complete the email and password fields as above.
2. Click **Sign In**.
3. A prompt will appear asking for your `Authenticator Code`.
4. Open the Google Authenticator app and copy the current 6-digit code for AmarVote.
5. Enter the code and click **Verify**.

> **Note**
>
> The TOTP code rotates every 30 seconds. If the code expires while you are typing, wait for the next code to appear and use that one.

3.3 Forgot Password

1. On the login page, click **Forgot Password?**.
2. Enter your registered email address and click **Send Reset Link**.
3. Check your email for a password reset link (valid for 15 minutes).
4. Click the link and set a new password.

Part II

Election Administration

Chapter 4

Creating an Election

This chapter is intended for Admins (and Users with election-creation permission). It walks through every option available when setting up a new election.

> **Important**
>
> The **Create Election** button will only appear in your navigation menu if you have been granted election-creation permission. If you cannot see it, contact your Owner.

4.1 Accessing the Election Creation Form

1. Log in to AmarVote.
2. In the left navigation sidebar, click **Create Election**.
3. The **New Election** form will open.

4.2 Basic Election Information

4.2.1 Election Name & Description

- `Election Name`: A short, descriptive title (e.g., "Student Council Election 2025").
- `Description`: A longer explanation of the election's purpose, visible to voters.

4.2.2 Visibility: Public vs. Private

Setting	Effect
Public	All registered users can see the election and its results.
Private	Only invited voters, assigned guardians, and the election admin can view the election, its ballot, and its results.

4.3 Voter Configuration

4.3.1 Open Voter List

If you want *any* registered AmarVote user to be able to vote:

1. Select **Anyone (All Registered Users)** from the `Voter List` dropdown.

4.3.2 Restricted Voter List (CSV Upload)

To limit voting to specific individuals:

1. Select **Upload Voter List** from the `Voter List` dropdown.
2. Prepare a CSV file with one column: `email`.
   Example: `voter1@example.com`, one per row.
3. Click **Upload Voters CSV** and select your file.
4. Review the voter count displayed after upload.

> **Note**
> Voters on the list who do not yet have an AmarVote account will need to register before they can vote. It is recommended to notify voters in advance.

4.4 Guardian Configuration

Guardians are trusted individuals who each hold a share of the cryptographic key used to decrypt the tally. Without a sufficient number (the *quorum*) of guardians cooperating, the results cannot be decrypted — ensuring no single party can tamper with or prematurely reveal results.

4.4.1 Uploading the Guardians List

1. Prepare a CSV file with one column: `email` (one guardian email per row).
2. Click **Upload Guardians CSV** and select your file.
3. The system will display the total number of guardians detected.

4.4.2 Setting the Quorum

The **quorum** is the minimum number of guardians who must cooperate to decrypt the tally. This is a critical security parameter.

1. In the `Quorum` field, enter a number between 1 and the total number of guardians.
2. Recommended: set quorum to at least $\lceil n/2 \rceil + 1$ where n is the total guardian count (a simple majority plus one).

Important

Quorum Example: If you have 5 guardians and set quorum to 3, any 3 guardians can decrypt the tally even if 2 guardians are unavailable. If quorum is set to 5, *all* guardians must be present — one absence blocks tallying permanently.

4.5 Ballot Configuration

4.5.1 Adding Candidates

For each candidate:

1. Click **Add Candidate**.
2. Enter the `Candidate Name`.
3. Enter the `Party Name` (optional).
4. Upload a `Candidate Photo` (optional, JPG/PNG, max 2 MB).
5. Upload a `Party Logo` (optional).
6. Repeat for each candidate.

Tip

Candidate photos and party logos significantly improve the voter experience, especially in large elections with many candidates. Providing them is strongly recommended.

4.5.2 Maximum Votes Per Voter

By default, each voter may select exactly one candidate. To allow multiple selections:

1. Locate the `Maximum Votes` field.
2. Enter the maximum number of candidates a voter may select (e.g., 3 means a voter can vote for 1, 2, or 3 candidates).

Note

Setting Maximum Votes to 1 creates a traditional single-choice election. Setting it to the total number of candidates allows approval voting.

4.6 Finalising Election Creation

After filling in all required fields:

1. Review the summary panel on the right side of the form.

2. Click **Create Election**.
3. The system will provision the election and redirect you to the election's management page.
4. The election will now show as **Setup Phase** — the guardian key ceremony must be completed before the election can go live.

Chapter 5

The Guardian Key Ceremony

The **Key Ceremony** is the cryptographic heart of AmarVote. It is a multi-phase process in which each guardian generates, shares, and verifies cryptographic key material. The purpose is to ensure that the election's decryption key is *never* held by any single person — it is split and distributed in a mathematically provable way.

> **Note**
>
> **Why does this matter?** Even the election administrator cannot decrypt the ballots alone. A minimum of *quorum* guardians must cooperate, making unilateral manipulation of results mathematically impossible.

5.1 Phase 1 — Key Pair Generation

5.1.1 Guardian Actions — Phase 1

Each guardian must complete Phase 1 independently:

1. Log in to AmarVote.
2. You will see a notification indicating you have been assigned as a guardian. Navigate to **All Elections** and find the election.
3. Open the election page and click the **Guardian** tab.
4. You will see that **Phase 1: Key Pair Generation** is active.
5. Click **Generate Key Pair**.
6. The system will cryptographically generate an **asymmetric key pair** (a public key and a private key) specifically for this election.
7. You will be prompted to create a **Guardian Password**. This password is used to encrypt your private key using AES-256 (a military-grade symmetric encryption standard).
8. **Important:** Click **Generate Secure Password** to let the system generate a random, strong password for you, *or* enter your own password in the field.
9. Click **Submit Key Pair**.

10. Your browser will automatically **download an encrypted private key file** (`.json` or `.enc`) to your computer.
11. Your **public key** is stored in the AmarVote database. Your **private key** is stored *only* on your device, encrypted with your guardian password.

Warning

Critical — Do not lose the following:

1. The downloaded private key file
2. Your guardian password

If you lose *either*, you will not be able to participate in decryption. Depending on the quorum, this could prevent the election results from ever being tallied.

Tip

Store the private key file in a secure location — an encrypted USB drive, a secure cloud vault, or your system's encrypted home directory. Store your guardian password in a password manager.

5.1.2 Phase Transition Condition

Phase 2 will not begin until **every** guardian assigned to the election has successfully completed Phase 1. The election page will show a progress indicator displaying how many guardians have submitted their public key.

5.2 Phase 2 — Backup Share Generation

Once all guardians have completed Phase 1, Phase 2 begins automatically. Each guardian must now generate **backup shares** — encrypted fragments of their private key that other guardians can use for disaster recovery.

5.2.1 How Backup Shares Work (Conceptual Overview)

Suppose there are 5 guardians (G1, G2, G3, G4, G5):

- Guardian G1 splits their private key into 4 fragments (one for each of the other 4 guardians).
- Each fragment is encrypted with the *public key* of the corresponding recipient guardian.
- For example, G1's fragment for G2 is encrypted so that *only G2* (using their private key) can decrypt it.
- These encrypted fragments are submitted to and stored in the database.

If G1 later loses their private key, any *quorum* number of other guardians can each provide their encrypted fragment. Together, the quorum can mathematically reconstruct G1's private key, allowing the tally to proceed.

5.2.2 Guardian Actions — Phase 2

1. Navigate to the election's **Guardian** tab.
2. You will see that **Phase 2: Backup Share Generation** is now active.
3. Click **Upload Encrypted Key File** and select the `.enc` file you downloaded in Phase 1.
4. Enter your `Guardian Password` in the provided field. The system will temporarily decrypt your private key in memory to generate the shares.
5. Click **Generate Backup Shares**.
6. The system will:
   (a) Decrypt your private key in memory.
   (b) Split it into $(n-1)$ encrypted fragments (one per other guardian).
   (c) Encrypt each fragment with the respective guardian's public key.
   (d) Submit all encrypted fragments to the server.
7. A confirmation message will appear when Phase 2 is complete for you.

> **Important**
> Your password and decrypted private key are used only in-browser during this step. They are **never transmitted** to the server in decrypted form. The server only receives the encrypted backup shares.

5.2.3 Phase 2 Transition Condition

Phase 3 begins only when **all** guardians have submitted their backup shares.

5.3 Phase 3 — Election Scheduling (Admin)

When all guardians have completed Phase 2, the admin is notified and Phase 3 becomes available.

5.3.1 Admin Actions — Phase 3

1. Log in as the election administrator.

2. Navigate to the election's management page.
3. You will see the **Phase 3: Schedule Election** panel is now active.
4. Select the `Election Start Date and Time`.
5. Select the `Election End Date and Time`.
6. Click **Publish Election**.
7. The election status changes to **Scheduled**. Voters will be able to vote starting from the specified start time.

Tip

Notify voters in advance of the election dates. If you uploaded a voter CSV, consider sending a separate reminder email to all voters with the election start time and access link.

Part III

The Voting Process

Chapter 6

Voting — Casting Your Encrypted Ballot

AmarVote uses an **end-to-end verifiable** voting protocol. Your ballot is encrypted *before* it is submitted to the server — the server never sees how you voted in plain text.

6.1 Accessing the Voting Booth

1. Log in to AmarVote during the election period.
2. Navigate to **All Elections** and find the election you wish to vote in.
3. Open the election page.
4. Click the **Voting Booth** tab.

> **Important**
>
> The **Voting Booth** tab will only appear if:
> - The election is currently open (between the start and end times).
> - Your email is on the approved voter list (or the election is open to all users).
> - You have not already cast your final ballot.

6.2 Selecting Your Candidate(s)

1. The voting booth displays all candidates with their names, party affiliations, and photos (if uploaded).
2. Select the candidate(s) you wish to vote for. The interface will show how many you may select (e.g., "Select up to 2 candidates").
3. A checkmark will appear next to each selected candidate.
4. Once you have made your selection(s), click **Create Encrypted Ballot**.

> **Note**
>
> **Important distinction:** Creating an encrypted ballot is *not* the same as casting your vote. This step only creates a cryptographic envelope for your choice. You

have two options from here: **cast** the ballot or **challenge** it (Benaloh Challenge).

6.3 The Benaloh Challenge

The **Benaloh Challenge** is a cryptographic audit mechanism that allows you to *verify* that the encrypted ballot accurately represents your choice — without decrypting and revealing your vote.

6.3.1 How It Works

When you create an encrypted ballot for candidates A and B:

1. The system generates an encrypted representation of your selection.
2. The challenge asks you to *re-enter* your intended vote selection.
3. The system checks whether the re-entered selection matches the encrypted ballot.
4. If it matches, you receive confirmation that the encryption is correct.
5. If it does not match, the encryption would be revealed as incorrect (which should never happen in a properly functioning system).

Note

Key property: If you voted for A and B, only submitting {A, B} in the challenge will return a positive result. Submitting {A}, {B}, {A, C}, etc., will return a negative result. This proves the encryption without revealing the actual vote to the server.

6.3.2 Performing the Benaloh Challenge

1. After creating an encrypted ballot, click **Challenge Ballot (Benaloh Challenge)**.
2. A selection interface will appear. Select what you believe you voted for.
3. Click **Submit Challenge**.
4. The system will report whether the challenge passed or failed.

Warning

Critical Rule: You may only perform the Benaloh Challenge **once** per encrypted ballot. Regardless of whether the challenge passes or fails, the challenged encrypted ballot is **permanently discarded**. You must then:

1. Re-select your candidates.
2. Create a new encrypted ballot.

3. Decide again: cast it or challenge it.

Tip

No limit on challenges: You may create an encrypted ballot, challenge it, discard it, and create another one as many times as you wish. This is by design — it allows you to verify the system's integrity without fear of running out of opportunities. However, you must eventually *cast* (not challenge) a ballot to have your vote counted.

6.4 Casting Your Vote

When you are satisfied that the system is working correctly and ready to cast your final vote:

1. Re-select your candidate(s) in the voting booth.
2. Click **Create Encrypted Ballot**.
3. Click **Cast Ballot** (do *not* click Challenge this time).
4. A confirmation dialog will appear. Review and click **Confirm & Cast Vote**.

Warning

This action is irreversible. Once you cast your ballot, it cannot be changed, recalled, or re-cast. Your vote is final.

6.5 Your Voter Receipt

After successfully casting your ballot, AmarVote will send you an email containing your **Voter Receipt**:

- **Ballot Tracking Code**: A unique identifier for your ballot in the tally system.
- **Ballot Hash**: A cryptographic fingerprint of your encrypted ballot.

Tip

Save your receipt! You will need both the tracking code and the hash to verify your vote after the tally is complete. Save the email or note these values in a secure place.

Part IV

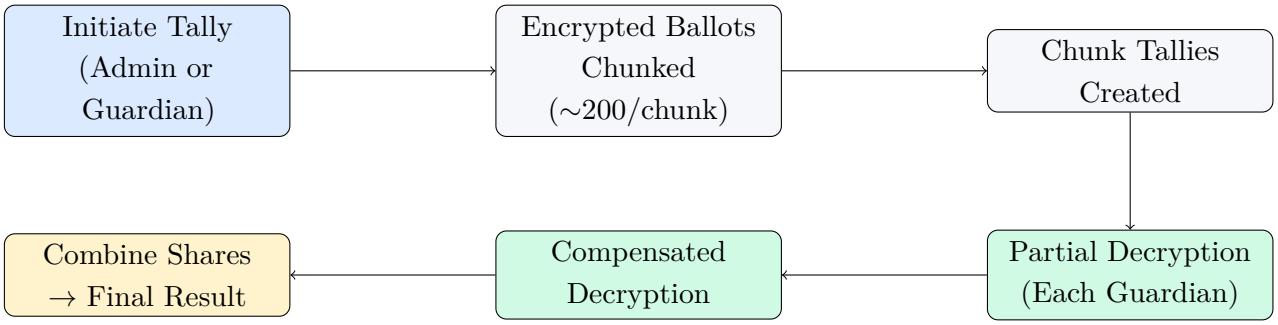
Tallying & Decryption

Chapter 7

The Tallying Process

After the election closes, the encrypted ballots must be tallied and then decryptted by the guardians to reveal the final results.

7.1 Overview of the Tally Pipeline



7.2 Initiating the Tally

Either the election admin or any guardian may start the tally process.

1. Navigate to the election page after the election has closed.
2. In the **Tally** or **Results** tab, click **Initiate Tally**.
3. The system will begin processing. This involves:
   - Retrieving all cast encrypted ballots from the database.
   - Dividing them into chunks of approximately 200 ballots each.
   - Computing an encrypted tally for each chunk using homomorphic addition.
   - Combining all chunk tallies into a single encrypted aggregate tally.
4. A progress indicator will display the number of chunks processed vs. total.
5. When complete, a **"Tally Complete"** confirmation message is displayed.

Note

The tally computation is performed asynchronously in the background. For large elections (thousands of votes), this may take several minutes. You do not need to remain on the page — progress is saved and you can return to check.

Chapter 8

Cryptographic Decryption

This is the final cryptographic phase. The encrypted tally can only be decrypted by combining a minimum of *quorum* guardian key shares.

8.1 Partial Decryption (Guardian Action)

1. Guardians are notified that the tally phase is complete and partial decryption can begin.
2. Each guardian navigates to the election page and opens the **Guardian** tab.
3. Click **Upload Encrypted Key File** and select your private key file from Phase 1.
4. Enter your `Guardian Password`.
5. Click **Submit Credentials & Decrypt**.
6. The system will:
   (a) Decrypt your private key in-browser using your password.
   (b) Apply your private key to each tally chunk — producing **partial decryptions** for your share.
   (c) Generate **compensated decryptions** for all other guardians (using the encrypted backup shares from Phase 2, decrypted with your private key). This allows the system to substitute for absent guardians.
   (d) Submit all partial and compensated decryptions to the server.
7. A confirmation message is displayed when your decryption contribution is submitted.

> **Important**
> The partial decryption process may take several minutes depending on the number of tally chunks. Do not close the browser tab while decryption is running.

8.2 Combining Decryption Shares (Admin or Guardian)

Once at least *quorum* guardians have submitted their partial decryptions:

1. Navigate to the election's **Tally** tab.
2. The **Combine Decryption Shares** button will become active.
3. Click **Combine Decryption Shares**.
4. The system mathematically combines all submitted partial and compensated decryption shares to produce the **final plaintext tally**.
5. The election results are now computed and published.

Tip

Results are displayed as a bar chart with vote counts per candidate, alongside percentage breakdowns and total ballot counts. All cryptographic proofs are also made available for public audit.

Part V

Verification & Audit

Chapter 9

Verifying Your Vote

AmarVote is designed to be **fully auditable**. Every voter, every observer, and every independent researcher can verify the integrity of the election.

9.1 Using Your Voter Receipt

After the results are published:

1. Navigate to the election page and open the **Verification** tab.
2. Click **Verify My Ballot**.
3. Enter your `Ballot Tracking Code` (from the receipt email).
4. Enter your `Ballot Hash` (from the receipt email).
5. Click **Verify**.

The system will confirm:

- **Ballot Found**: Your tracking code exists in the tally, confirming your ballot was *recorded*.
- **Hash Match**: The ballot hash in the tally matches your receipt, confirming your ballot was *not modified* after submission.

> **Tip**
>
> If the ballot hash does not match, this indicates potential tampering. Report this immediately to the election administrator and your national or institutional election authority.

9.2 Manual Ballot Lookup

If you prefer to verify without relying on the automated tool:

1. Navigate to the election's **Ballots in Tally** tab.
2. This displays the complete list of all encrypted ballots included in the tally.
3. Manually search through the list for your Ballot Tracking Code.

4. Compare the displayed ballot hash with the hash in your receipt email.

> **Note**
> This manual approach is deliberately available for maximum transparency. Ballot tracking codes are anonymised — they identify the ballot but not the voter — so this list is safe to make public.

9.3 Public Election Data Download

All election data (except private keys) is publicly available for independent audit:

1. Navigate to the election's **Verification** tab.
2. Click **Download Election Data**.
3. The download package includes:
   - All encrypted ballots
   - Public keys of all guardians
   - The encrypted tally
   - All partial decryptions and proofs
   - The final plaintext results

> **Note**
> Any technically capable third party can use this downloaded data to re-verify the entire election independently using the ElectionGuard specification. This is the gold standard of election transparency.

Appendix A

Frequently Asked Questions

Can the administrator see how I voted?

No. Your ballot is encrypted using the combined public keys of all guardians before it is submitted. The administrator cannot decrypt individual ballots — only the aggregate tally can be decrypted, and only with guardian cooperation.

What if a guardian is unavailable during decryption?

If the number of available guardians meets or exceeds the quorum threshold, the election can still be tallied. The backup shares generated during Phase 2 are used to compensate for absent guardians.

Can I vote more than once?

No. The system enforces a one-final-ballot-per-voter policy. While you can create and challenge multiple encrypted ballots (for verification purposes), casting your vote is a one-time action.

Is my identity linked to my ballot?

Your voter receipt email contains a Ballot Tracking Code and Hash. These identifiers are mathematically derived from the *encrypted* ballot — they do not reveal your identity or your vote. The public tally list contains tracking codes but no voter names.

What is the Benaloh Challenge for?

It is a cryptographic proof that the encryption software is working correctly — that the encrypted ballot represents what you actually selected. It provides mathematical confidence without revealing your vote to the server.

What if I lose my guardian credentials?

Immediately notify the election administrator. If the remaining guardians still meet the quorum threshold, tallying can proceed without you (using compensated decryption). If your loss would drop available guardians below quorum, the election results may be unrecoverable.

Appendix B

Security Architecture Summary

Component	Security Property
Ballot encryption	ElectionGuard homomorphic encryption (El Gamal based)
Private key storage	AES-256 encryption with guardian-chosen password
Key distribution	Shamir's Secret Sharing (threshold scheme)
Ballot verification	Zero-knowledge proofs included with each ballot
Guardian backup	Encrypted with each guardian's public key
Transit security	HTTPS/TLS (Nginx production configuration)
Authentication	JWT tokens + optional TOTP 2FA
Password hashing	bcrypt (Spring Security default)
Infrastructure	Docker containers, PostgreSQL, Redis, RabbitMQ